PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 1 • Chapter OL-T49 Classified

Cross-session • chapter-organised candidate

11 classified items • 53 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Unit 1 • Chapter OL-T49 • 11 items • 53 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Sine, Cosine Rule & Area of Triangles

Unit 1 • 53 marks • 11 item(s)

June 2025 | 4MA1-1H Q20 | 5 marks
Use two non-right triangles sharing a diagonal

June 2024 | 4MA1-1H Q19 | 5 marks
Use two non-right triangles sharing a diagonal

November 2023 | 4MA1-1H Q10 | 5 marks
Choose between sine and cosine rule from data pattern

June 2023 | 4MA1-2H Q22 | 5 marks
Combine non-right trigonometry with circle or elevation geometry

June 2022 | 4MA1-1H Q18 | 5 marks
Find an acute angle using the sine rule

January 2022 | 4MA1-2H Q23 | 6 marks
Resolve the ambiguous sine-rule angle

January 2020 | 4MA1-1H Q14 | 3 marks
Use cosine rule inside a quadrilateral or composite shape

June 2021 | 4MA1-2H Q18 | 6 marks
Combine non-right trigonometry with circle or elevation geometry

November 2021 | 4MA1-2H Q18 | 5 marks
Use two non-right triangles sharing a diagonal

January 2023 | 1H Q16 | 5 marks
Find a side using the sine rule

November 2025 | 1H Q19 | 3 marks
Recover a side or angle from triangle area

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 13

4MA1-1H Q20

ORIGINAL QUESTION
5 marks

20

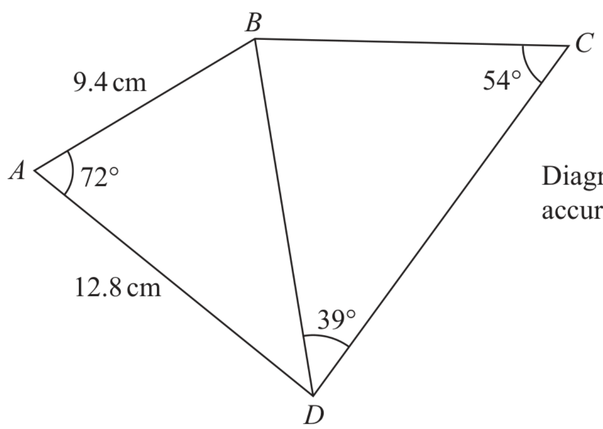

Diagram **NOT** accurately drawn

Work out the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 5 marks)

Worked Solution 13

Cosine rule followed by sine rule

UNIT 1
5 marks

Find the shared side

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ, \quad BD = 13.335597\ldots$$

Use the sine rule

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ} \Rightarrow BC = \frac{13.335597\ldots \sin 39^\circ}{\sin 54^\circ} = 10.3735\ldots$$

Final answer

$$BC = 10.4 \text{ cm } (3 \text{ s.f.})$$

Dependency check Retain the unrounded value of BD for the second triangle.

Question 13

4MA1-1H Q19

ORIGINAL QUESTION

5 marks

19 Here is quadrilateral *ABCD*

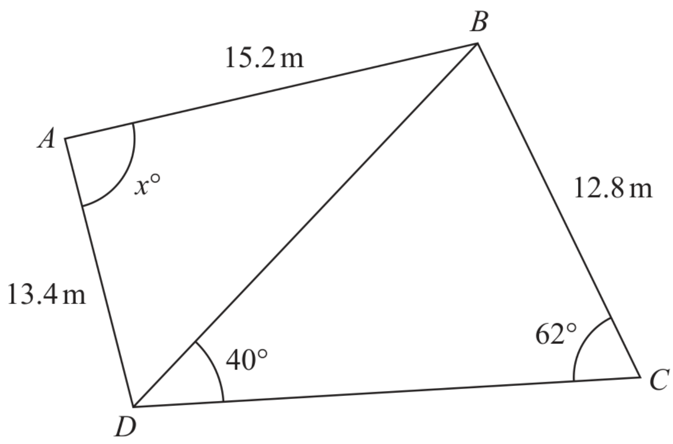

Diagram **NOT** accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$ ....

Worked Solution 13

Sine rule and cosine rule

MODULAR UNIT 1

5 marks

Diagonal

$$BD = 12.8 \sin 62^\circ / \sin 40^\circ = 17.582\ldots$$

Required angle

$$\cos x = \frac{13.4^2 + 15.2^2 - BD^2}{2(13.4)(15.2)}.$$

Final answer

$$75.6^\circ$$

Question 09

4MA1-1H Q10 demand

ORIGINAL QUESTION UNIT 1

5 marks

10 Here is triangle *ABC*

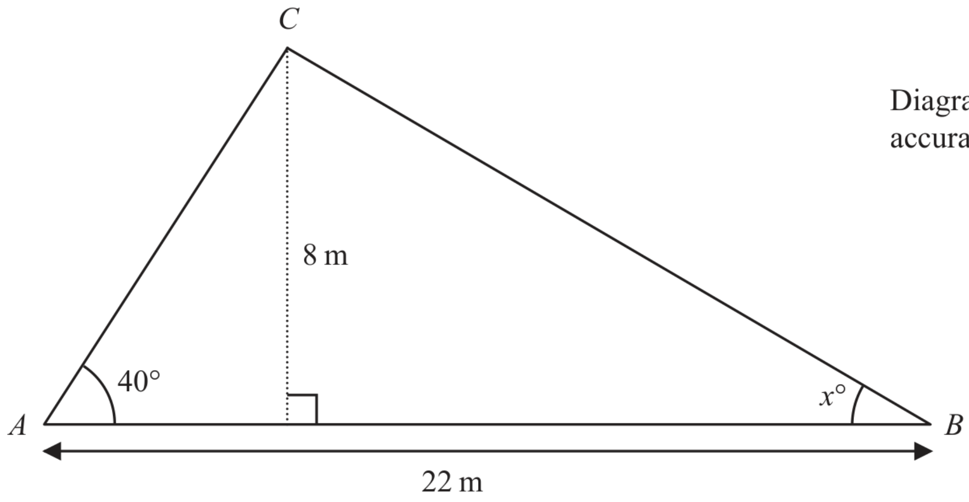

Diagram **NOT** accurately drawn

Work out the value of x
Give your answer correct to one decimal place.
Show your working clearly.

$x =$ ....

Worked solution 09

Angle in a triangle from a perpendicular height

MODULAR UNIT 1
5 marks

Demand – Angle in a triangle from a perpendicular height

Find the left base segment

$$AD = \frac{8}{\tan 40^\circ} = 9.534...$$

Find the right base segment

$$DB = 22 - 9.534... = 12.466...$$

Use the right triangle at B

$$\tan x = \frac{8}{12.466...}.$$

Evaluate

$$x = 32.690...^\circ \approx 32.7^\circ.$$

Final answer

32.7°

Question 26

4MA1-2H Q22

ORIGINAL QUESTION
5 marks

22 The diagram shows a triangle *ABC* and a flagpole *BF*

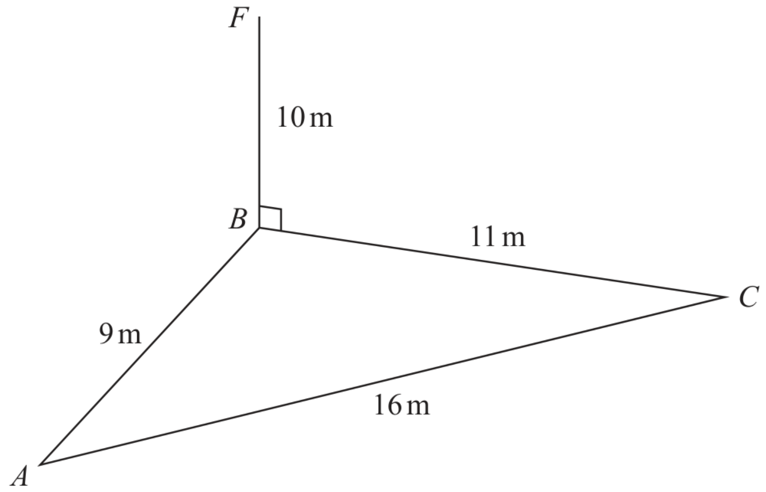

Diagram **NOT** accurately drawn

A, *B* and *C* are points on horizontal ground.

BF is vertical.

$AB = 9\text{ m} \qquad BC = 11\text{ m} \qquad AC = 16\text{ m} \qquad BF = 10\text{ m}$

D is the point on *AC* such that angle $BDC = 90°$

Work out the size of the angle of elevation of the point *F* from the point *D*
Give your answer correct to one decimal place.

.... °

Worked Solution 26

Angle of elevation from a perpendicular foot

MODULAR UNIT 1

5 marks

Find the ground angle

$$\cos\angle BCA = \frac{11^2 + 16^2 - 9^2}{2(11)(16)} = \frac{37}{44} \Rightarrow \angle BCA \approx 32.763^\circ.$$

Find BD

$$BD = 11\sin 32.763^\circ \approx 5.95\ \text{m}.$$

Use the vertical flagpole

$$\tan\angle FDB = \frac{BF}{BD} = \frac{10}{5.95}.$$

Evaluate

$$\angle FDB \approx 59.2^\circ.$$

Final answer

$$59.2^\circ$$

Original question 12

4MA1-1H Q18 Q18

ORIGINAL QUESTION
5 marks

18 Here is triangle ABC

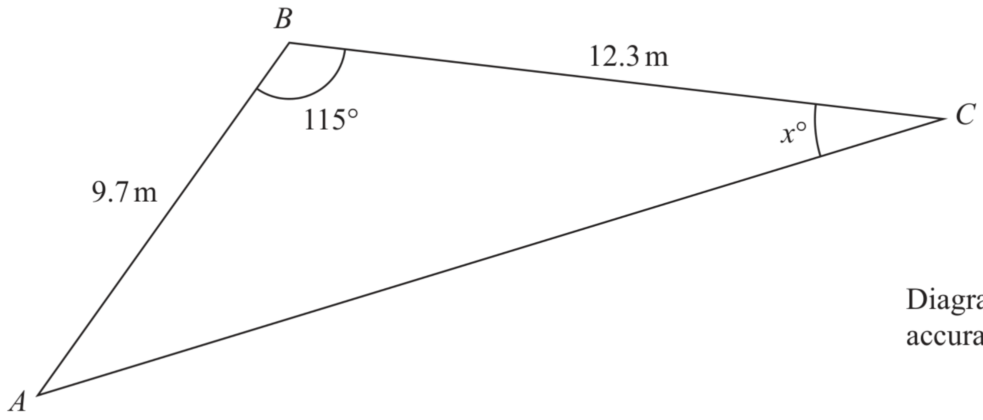

Diagram **NOT** accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$ ....

Worked solution 12

Chapter: Sine, Cosine Rule & Area of Triangles

MODULAR UNIT 1
5 marks

Q18 – Chapter: Sine, Cosine Rule & Area of Triangles

Family: Use the sine rule • Variant: Find an acute angle using the sine rule

Method
By the cosine rule, AC^2 = 9.7^2 + 12.3^2 - 2(9.7)(12.3)cos(115 degrees).
Why: The two known sides enclose the known 115-degree angle.

Method
AC^2 = 346.2... , so AC = 18.606... m.
Why: Evaluate the cosine-rule expression without premature rounding.

Method
Using the sine rule, sin(x)/9.7 = sin(115 degrees)/18.606... .
Why: Side AB = 9.7 is opposite angle x, and AC is opposite 115 degrees.

Method
Therefore sin(x) = 9.7 sin(115 degrees)/18.606... = 0.472... .
Why: Rearrange the sine-rule equation for sin(x).

Method
x = sin^-1(0.472...) = 28.2 degrees to 3 significant figures.
Why: Take the inverse sine in degree mode and round once at the end.

Final answer
angle: x = 28.2 degrees

Original question 27

4MA1-2H Q23

ORIGINAL QUESTION
6 marks

23 The diagram shows triangle *PQR*

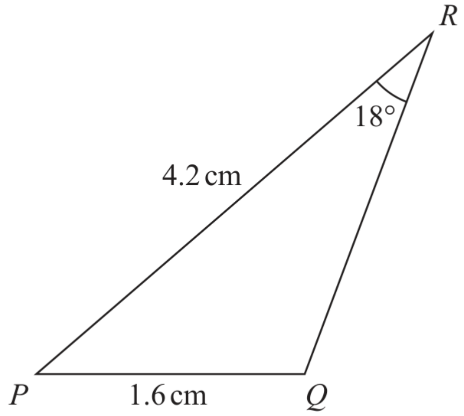

Diagram **NOT** accurately drawn

$PQ = 1.6\text{ cm}$ $PR = 4.2\text{ cm}$ Angle $PRQ = 18°$

Given that angle *PQR* is obtuse,

work out the area of triangle *PQR*
Give your answer correct to 3 significant figures.

.... cm^2

Worked solution 27

Chapter: Sine, Cosine Rule & Area of Triangles

MODULAR UNIT 1
6 marks

Q23 – Chapter: Sine, Cosine Rule & Area of Triangles

Family: Use the sine rule • Variant: Resolve the ambiguous sine-rule angle

Method
Using the sine rule, $\sin Q/4.2=\sin 18°/1.6$, giving $Q=54.2°$ or $125.8°$.
Why: The sine equation has an acute and an obtuse possibility.

Method
The question says Q is obtuse, so $Q=125.8°$.
Why: Select the permitted branch.

Method
$P=180-125.8-18=36.2°$.
Why: Angles in a triangle total 180°.

Method
The corresponding side RQ is 3.0585... cm.
Why: Use the cosine rule or the sine rule after finding P.

Method
$\text{Area}=1/2\times4.2\times1.6\times\sin 36.2°$.
Why: Use two sides and the included angle.

Method
$\text{Area}\approx1.98\text{ cm}^2$ (3 s.f.).
Why: Round the calculator value.

Final answer

area: 1.98 cm^2

14 The diagram shows parallelogram *EFGH*.

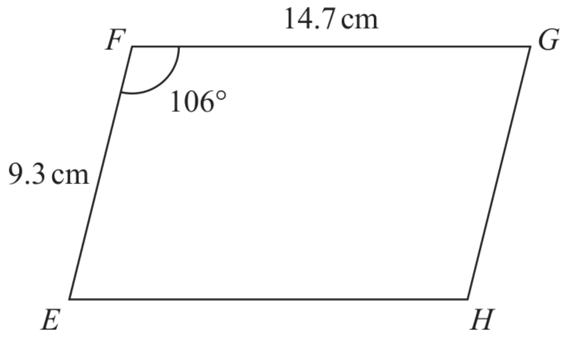

Diagram **NOT** accurately drawn

$EF = 9.3$ cm
$FG = 14.7$ cm
Angle $EFG = 106°$

(a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

(b) Work out the length of the diagonal *EG* of the parallelogram.
Give your answer correct to 3 significant figures.

.... cm
(3)

Worked solution

January 2020 | Question 14 | 3 marks

Family: Use the cosine rule • Variant: Use cosine rule inside a quadrilateral or composite shape

Method / working

1. $EG^2=9.3^2+14.7^2-2(9.3)(14.7)\cos 106°$.
2. $EG=19.440\ldots$ cm.
3. To 3 s.f., $EG=19.4$ cm.

Final answer: 19.4 cm

18

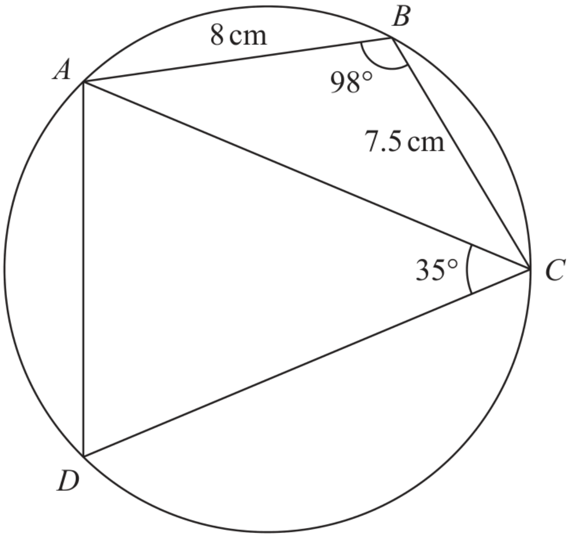

Diagram **NOT** accurately drawn

ABCD is a quadrilateral where *A*, *B*, *C* and *D* are points on a circle.

$AB = 8$ cm
$BC = 7.5$ cm
Angle $ABC = 98°$
Angle $ACD = 35°$

Work out the perimeter of quadrilateral *ABCD*.
Give your answer correct to one decimal place.

.... cm

Worked solution

June 2021 | Question 18 | 6 marks

Family: Resolve mixed non-right-triangle chains • Variant: Combine non-right trigonometry with circle or elevation geometry

Method / working

1. Cyclic quadrilateral: angleADC=180-98=82°.
2. In triangle ADC, angleDAC=180-82-35=63°.
3. Use the cosine rule in triangle ABC to find AC~11.7 cm.
4. Use the sine rule: AD=AC·sin35°/sin82°~6.77 cm.
5. Use the sine rule: DC=AC·sin63°/sin82°~10.5 cm, then add 8+7.5+AD+DC.
6. Round the perimeter to 1 decimal place: 32.8 cm.

Final answer: 32.8 cm

18 Here is a quadrilateral *ABCD*.

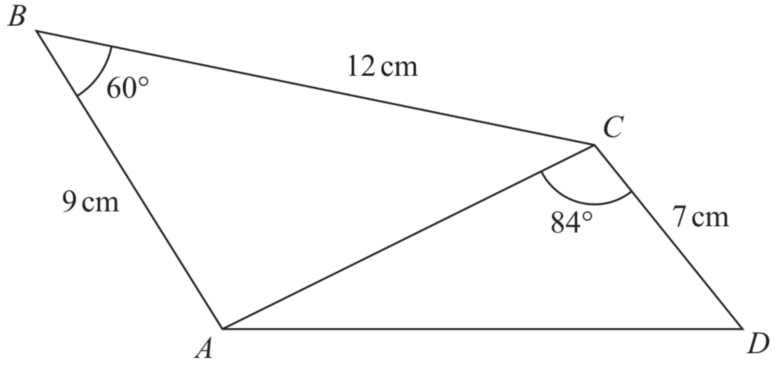

Diagram **NOT** accurately drawn

Calculate the area of quadrilateral *ABCD*.
Give your answer correct to 3 significant figures.
Show your working clearly.

.... cm^2

Worked solution

November 2021 | Question 18 | 5 marks

Family: Resolve mixed non-right-triangle chains • Variant: Use two non-right triangles sharing a diagonal

Method / working

1. Area of triangle ABC:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. $\frac12(9)(12)\sin 60^\circ=46.765\ldots$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. Find AC using the cosine rule:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. $AC^2=9^2+12^2-2(9)(12)\cos 60^\circ$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. $AC=10.816\ldots$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. Area of triangle ACD:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. $\frac12(10.816\ldots)(7)\sin 84^\circ=37.642\ldots$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. Total area:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
9. $46.765\ldots+37.642\ldots=84.407\ldots$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
10. Answer: $84.4 cm^2$.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: 84.4 cm^2.

16 Here is a shape formed from two triangles ABC and CDE
ACD and BCE are straight lines.

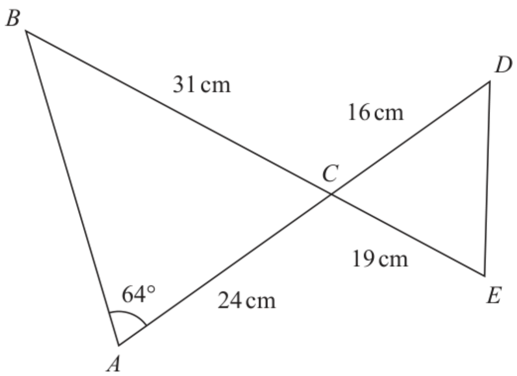

Diagram **NOT** accurately drawn

$AC = 24\text{ cm}$ $BC = 31\text{ cm}$ $CE = 19\text{ cm}$ $CD = 16\text{ cm}$

Angle $BAC = 64°$

Work out the length of DE
Give your answer correct to 3 significant figures.

January 2023 | 1H | Question 16 | 5 marks

Chapter OL-T49: Sine, Cosine Rule & Area of Triangles

.... cm

(Total for Question 16 is 5 marks)

Answer reference | January 2023 | Question 16

Family: Use the sine rule • Variant: Find a side using the sine rule

16	$\frac{\sin ABC}{24}=\frac{\sin 64}{31}$ oe		5	M1
	$(ABC=)\sin^{-1}\left(\frac{24\times\sin 64}{31}\right)(=44.\ldots)$			M1
	180 – "44.…" – 64 (= 71.9…)			M1 accept 72
	$(DE^2=)16^2+19^2-2\times16\times19\times\cos"71.9..."$ or $(DE=)\sqrt{16^2+19^2-2\times16\times19\times\cos"71.9..."}$ or $(DE=)\sqrt{617-181.8...}$ or $\sqrt{428.166...}$			M1 for DE^2 or DE
	Correct answer scores full marks (unless from obvious incorrect working)	20.7		A1 awrt 20.7
				Total 5 marks

Worked solution

January 2023 | Question 16 | 5 marks

Family: Use the sine rule • Variant: Find a side using the sine rule

Official final mark-scheme pages are embedded immediately after this question.

19 The diagram shows triangle *ABC*

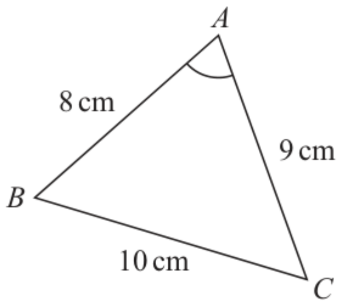

Diagram **NOT** accurately drawn

$AB = 8\text{ cm}$ $BC = 10\text{ cm}$ $CA = 9\text{ cm}$

Work out the size of angle *BAC*
Give your answer correct to one decimal place.

.... °

(Total for Question 19 is 3 marks)

Worked solution

November 2025 | Question 19 | 3 marks

Family: Use area equals one-half ab sine C • Variant: Recover a side or angle from triangle area

Solution

Use the cosine rule for angle BAC:

cos A=(8^2+9^2-10^2)/(2(8)(9))

cos A=(45)/(144)

A=71.790ldots^circ

Answer: 71.8^circ.

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION